MPA – TVN High Voltage Technology

Brno University of Technology

Measurement of partial discharges with a coupling capacitor.

Semester Project

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Abstract - **Partial discharge (PD) measurement is a paramount diagnostic procedure for assessing the insulation integrity of high-voltage (HV) equipment, including power cables, rotating machines, and switchgear. As insulation systems age under thermal, electrical, mechanical, and environmental stresses, defects such as voids, de laminations, and surface tracking can develop, leading to catastrophic failure if undetected. This project presents a comprehensive analysis of PD detection using capacitive couplers, a method favoured for its sensitivity and ability to capture high-frequency phenomena.**

The study begins by reviewing the fundamental physics of partial discharges, contrasting the classical capacitive equivalent circuit (''abc" model) with the more physically rigorous dipole model proposed by Pedersen. It explores the design and bandwidth characteristics of capacitive sensors, comparing simple RC topologies with high-performance broadband double-T filters capable of minimizing signal distortion. Furthermore, the project examines the practical application of these couplers in both on-line and off-line testing environments. Critical aspects such as signal attenuation in lossy transmission lines (cables), interference suppression using balanced bridge techniques, and the interpretation of Phase Resolved Partial Discharge (PRPD) patterns are discussed in detail. The analysis confirms that while capacitive coupling is a mature technology, its efficacy relies heavily on correct impedance matching, bandwidth selection, and advanced signal processing to discriminate between noise and true defect signals.

1 Introduction

High-voltage insulation systems constitute the backbone of the electrical power grid. Whether in the form of cross-linked polyethylene (XLPE) in power cables, epoxy-mica systems in rotating machines, or oil-paper insulation in transformers, these materials are subjected to continuous operational stresses. Among the various degradation mechanisms, Partial Discharge (PD) is both a symptom and a cause of insulation failure. Defined by IEEE Std 1434-2014 as an electrical discharge that only partially bridges the insulation between conductors, PD can occur internally within voids, on the surface of dielectric materials, or at electrode interfaces [1].

The detection of PD is crucial for condition-based maintenance (CBM). While acoustic, optical, and chemical methods exist, electrical pulse sensing remains the industry standard for quantification. This method relies on the fact that a PD event involves a rapid movement of charge, resulting in a current pulse that propagates through the system.

1.1 Diagnostic Methods

Various methodologies have been developed to detect PD activity, categorized by the physical phenomenon they measure:

- **Acoustic Detection**: Measures the ultrasonic shock waves generated by the discharge. While useful for localization, it is subject to high attenuation within the equipment.
- **Chemical Detection**: Monitors byproducts such as ozone (O3) or dissolved gases in oil (DGA). This is generally a slow, cumulative indicator rather than a real-time diagnostic.
- Electrical Detection: Measures the transient current pulses generated by the movement of charge during a discharge event. This method allows for the quantification of the discharge magnitude in (*pC*) and provides high temporal resolution.

1.2 Scope of Project

This project focuses specifically on the **Coupling Capacitor** method. A coupling capacitor acts as a high-pass filter, blocking the high-voltage power frequency (50 or 60 Hz) while providing a low-impedance path for the high-frequency (HF) or very-high-frequency (VHF) transient signals generated by the discharge. The objective of this work is to provide a holistic overview of this technology, ranging from the theoretical electrostatics of the discharge event to the practical engineering challenges of sensor installation and signal analysis.

2 Theoretical Background and PD Models

To design effective detection systems and interpret the resulting data, a robust physical model of the partial discharge event is required. Two primary models dominate the literature: the classical capacitive equivalent circuit and the field theoretical dipole model.

2.1 The Capacitive Equivalent Circuit (The ‘’abc” Model)

Historically, the ‘’abc” model has been the pedagogical standard for explaining PD phenomena. As critically reviewed by Lemke [2], this model abstracts the complex geometry of an insulation defect into a simplified circuit of three capacitors:

- C_a: Represents the capacitance of the bulk, healthy dielectric material.
- C_b: Represents the capacitance of the healthy dielectric in series with the defect.
- C_c: Represents the capacitance of the void or cavity itself.

In this model, a partial discharge occurs when the voltage across the cavity, V_c, exceeds the breakdown strength of the gas trapped within. This is simulated by the closing of a switch across C_c, which causes a rapid discharge of the cavity capacitance. This local discharge results in a voltage drop across the terminals of the test object, which causes a redistribution of charge from the external circuit.

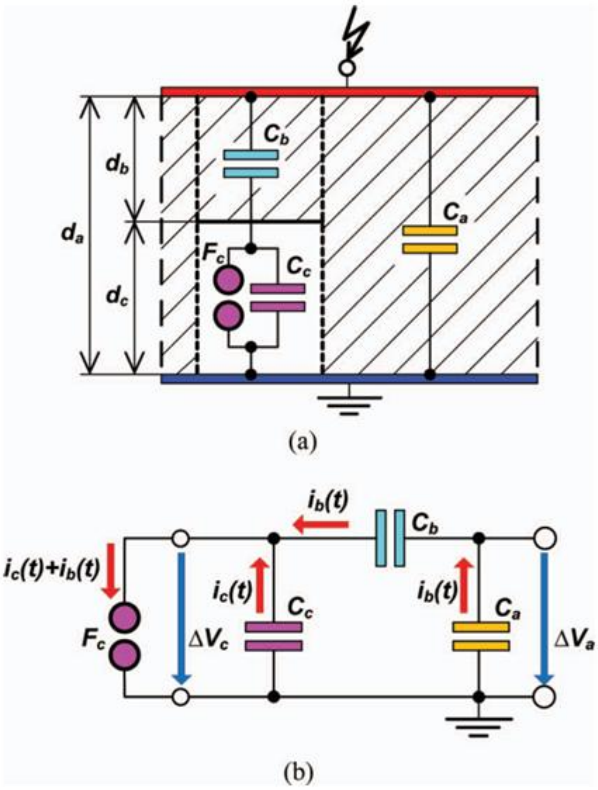

Figure 1The classical capacitive ‘’abc” equivalent circuit model for partial discharge [2].

The discharge of C_c causes a momentary voltage dip across the entire test object. To restore the voltage, charge flows from the external circuit (and the coupling capacitor). This measurable charge is termed the **Apparent Charge** (q_{app}). It is crucial to note that q_{app} is not the actual charge transferred across the void (q_c). The relationship is governed by the capacitive divider formed by the healthy insulation:

$$q_{app} \approx q_c \cdot \frac{Cb}{Ca+Cb}$$

Since the bulk capacitance C_a is typically orders of magnitude larger than the series capacitance C_b (because the volume of healthy insulation is vast compared to the series channel), the equation simplifies to:

$$q_{app} \approx q_c \cdot \frac{Cb}{Cc}$$

This relationship highlights a fundamental limitation: we never measure the true discharge magnitude directly, but rather its effect on the external terminals.

2.2 The Dipole Model

While the ''abc" model is useful for circuit analysis, it lacks physical rigor regarding the actual movement of charge carriers. A more accurate approach, initially proposed by Pedersen and discussed by Lemke [2], is the **Dipole Model**. This model rejects the notion of ''capacitance" for a void, arguing that capacitance is a property defined by electrodes, which a void typically lacks. Instead, the PD event is modelled as a sudden change in the polarization vector P within the dielectric volume. When charge carriers (electrons and positive ions) move across a void during a discharge, they create a transient electric dipole moment.

The signal detected at the sensor terminals is the **induced charge** resulting from this change in the internal electric field distribution. According to the Ramo Shockley theorem applied to dielectrics, the induced current *i(t)* at the terminals is proportional to the velocity of the charge carriers v and the coupling weighting function $\nabla\ \lambda$:

$$i(t) = q \cdot v \cdot \nabla \lambda$$

Here, is a dimensionless scalar field that represents the coupling efficiency between the location of the charge and the sensing electrode. The Dipole Model explains why the orientation of the discharge relative to the electric field and the position of the sensor critical factors in sensitivity are. It confirms that what we measure is an induced phenomenon, distinct from the physical charge transport inside the void.

The Dipole Model provides a better explanation for why the geometry of the sensor and its location relative to the defect significantly influence sensitivity.

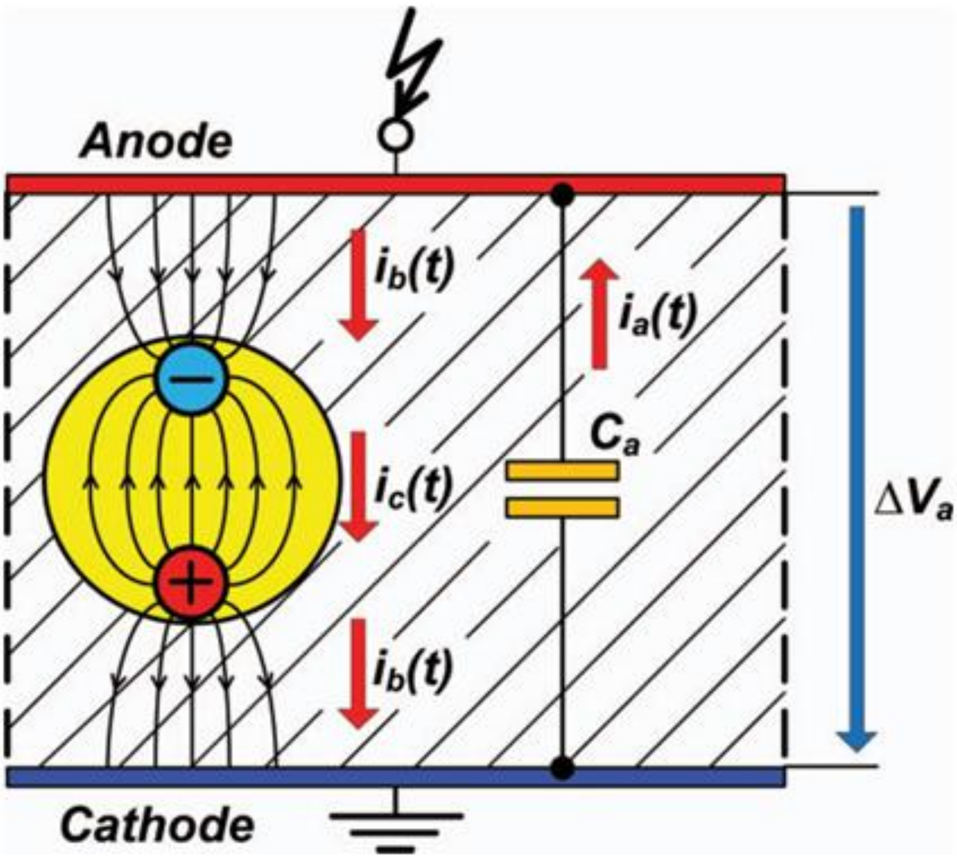

Figure 2 The field-theoretical Dipole Model representing PD as a change in dipole moment [2].

3 The Capacitive Coupler: Sensor Design and Topologies

The capacitive coupler is the primary interface between the high-voltage physics occurring in the equipment and the low-voltage electronics of the measuring instrument. Ideally, a coupler should withstand the full line voltage, have negligible partial discharge of its own (PD-free), and offer a flat frequency response over the bandwidth of interest.

3.1 Basic RC Topology

The most fundamental coupler design consists of a high-voltage capacitor (C_K) connected in series with a detection impedance (Z_m), typically a resistor or an inductor-capacitor (RLC) circuit. As described by Zhong and Xu [3], typical values for coupling capacitors range from 10 pF to several nanofarads (nF), depending on the voltage class and the required sensitivity. The frequency response of this simple RC circuit is that of a high-pass filter. The cut-off frequency fc is determined by:

$$f_c = \frac{1}{2\pi RC}$$

For a standard detection impedance of 50 and a 1000 *pF* coupling capacitor, the cut-off frequency is approximately 3 *MHz*. This topology is simple and robust but has limitations. Because it essentially differentiates the signal (outputting di/dt rather than $i(t)$), it can distort the pulse shape, making advanced analysis techniques like pulse-shape classification difficult. Additionally, the limited bandwidth may miss the lower frequency components of the discharge, leading to integration errors when calculating charge.

3.2 High-Performance Broadband Couplers

Modern diagnostic systems often require bandwidth extending from a few kHz up to tens of MHz to capture the full fidelity of the PD pulse. To achieve this, Rodrigo et al. [4] proposed a ‘‘double T” filter topology. This advanced design utilizes multiple stages of high-voltage capacitors combined with active operational amplifiers. The advantage of the broadband coupler is its flat frequency response. Unlike the simple RC coupler which differentiates the signal (outputting the derivative of the current), the broadband coupler can be designed to output a signal proportional to the current itself. This allows for accurate integration to determine the charge:

$$Q = \int i(t)\, dt$$

Rodrigo's research indicates that this topology reduces measurement uncertainty significantly. In comparative tests, the broadband coupler achieved charge evaluation uncertainty of approximately **1.10%,** whereas standard RC couplers operating with limited bandwidth (100 kHz cut-off) exhibited uncertainties nearing **7%** [4]. This precision is vital for correlating PD magnitude with insulation damage.

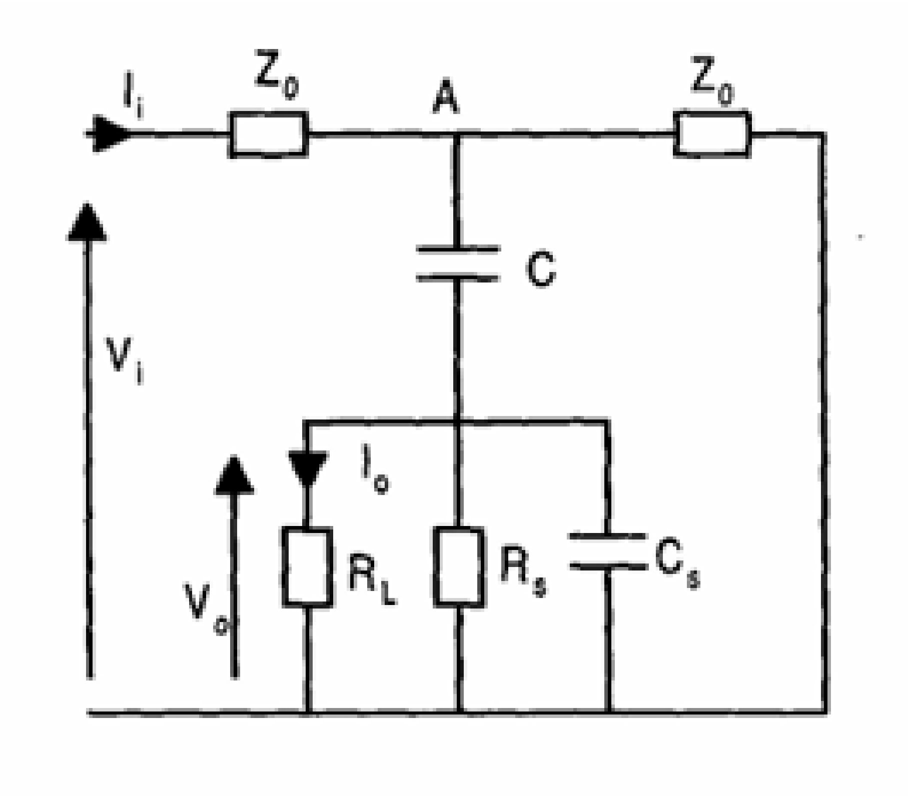

Figure 3 Equivalent circuit diagram of a basic capacitive coupler [3].

3.3 Coupler Specifications

According to IEEE Std 1434 [1], coupling capacitors used for on-line monitoring must meet stringent safety and performance criteria:

- **Voltage Rating**: Must withstand at least 2.17 times the rated line-to-line voltage during voltage endurance testing.
- **PD free**: The coupler itself must have a PD extinction voltage higher than the system operating voltage.

4 Measurement Configurations and Interference Suppression

In practical environments, especially on-site substations, PD signals are often buried in electrical noise (switching operations, radio broadcasts, power electronics). The configurationofthecouplingcapacitorsisthereforeessentialforachievingawork able Signal-to-Noise Ratio (SNR).

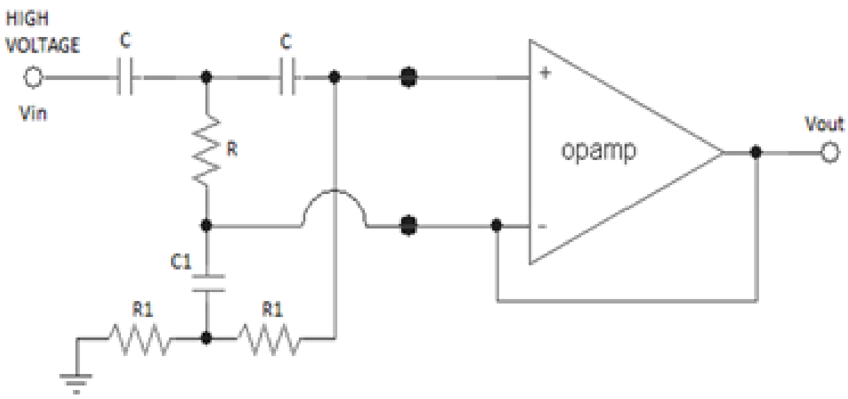

Figure 4 Schematic of a high-performance broad band capacitive coupler [4].

4.1 Balanced Detection Circuits

For testing large capacitive objects, such as collective capacitor banks, the background noise can be overwhelming. Feng et al. [5] describe a balanced bridge detection circuit. This configuration utilizes two parallel arms:

1. The test object branch (C_x).
2. A balancing branch (reference capacitor or identical test object).

Both arms are connected to adjustable compensation reactors (L). The out puts from both arms are fed into a differential measuring unit. The principle of operation relies on **Common Mode Rejection**. External noise traveling down the power lines enters both arms of the bridge with equal magnitude and phase. The differential amplifier subtracts these signals, cancelling the noise. However, a partial discharge event originates inside the test object (C_x) and appears only in that arm (or with different polarity in the reference arm), and is thus preserved and amplified. Using this balanced method, Feng et al. demonstrated the ability to suppress background noise from over 200 pC down to below ***23pC*** in a live 220 kV substation environment, enabling the detection of subtle internal defects [5].

4.2 Signal Attenuation in Power Cables

When measuring PD in long power cables, the cable itself acts as a lossy transmission line. The high-frequency components of the PD pulse are attenuated much more rapidly than low-frequency components as they propagate away from the source. Tian et al. [6] demonstrated that this attenuation is frequency-dependent; the semiconducting screens of the cable act as a low-pass filter, aggressively damping VHF/UHF components while allowing lower frequencies to pass.

To accurately locate defects in cables, **Time of Flight** (*ToF*) analysis is employed. By installing capacitive couplers at two ends of the cable (or at a joint and a terminal), the location of the defect (x) can be calculated based on the time difference (△t) of pulse arrival:

$$x = \frac{L - v \cdot \triangle t}{2}$$

- L is the total cable length.
- v is the propagation velocity of the signal in the cable dielectric (typically 150 170 m/ μs for XLPE).
- △t is the time difference of arrival between the two sensors.

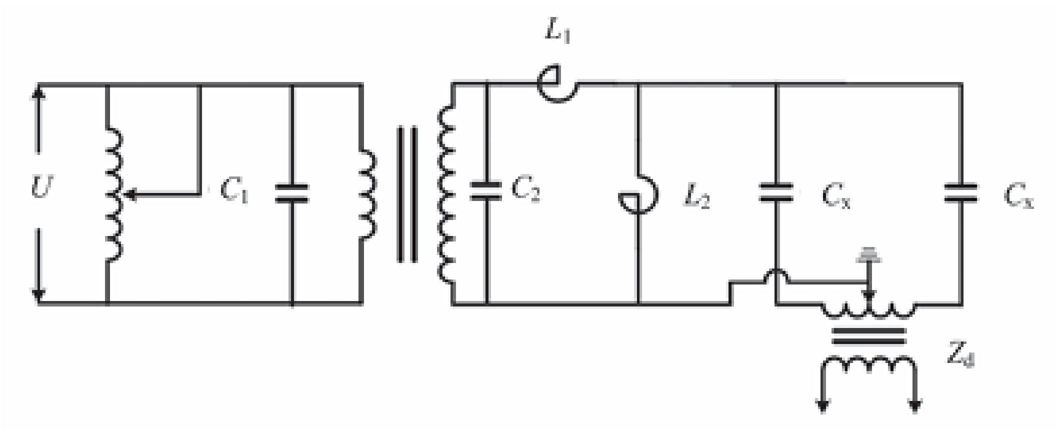

Figure 5 Balanced detection circuit for noise suppression [5].

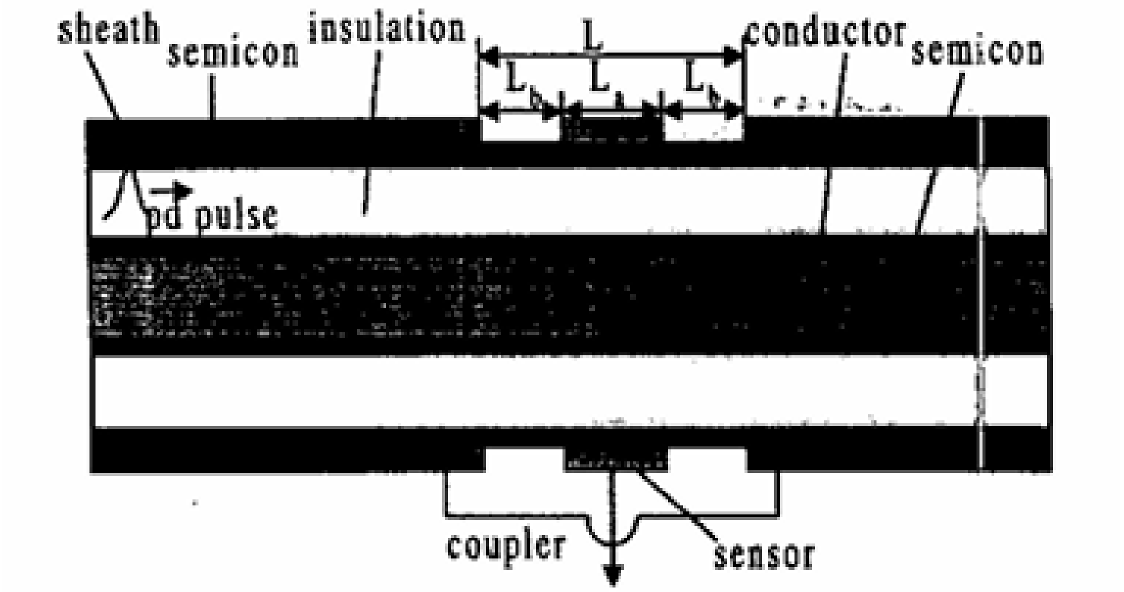

Figure 6 Installation of a capacitive coupler on a cable joint for signal detection [6].

5 Signal Analysis and Pattern Recognition

Acquiring the raw signal is only the first step. The core of PD diagnostics lies in identifying the **type** of defect (e.g., void, corona, surface tracking) based on the signal characteristics. The industry-standard method for this is **Phase Resolved Partial Discharge (PRPD)** analysis, where individual discharge pulses are plotted against the phase angle of the applied AC voltage.

5.1 Internal Discharge (Voids)

Internal discharges occur within gas-filled cavities enclosed in the insulation (e.g., bubbles in epoxy or delamination in mica tape). According to experiments by Chou and Chen [7], these discharges are characterized by a symmetrical pattern. In a PRPD plot, internal discharges typically appearing the **first ($0° - 90°$)**and **third ($180° - 270°$)** quadrants of the AC voltage cycle. These correspond to the rising edges of the positive and negative half-cycles, where the voltage across the void is increasing most rapidly. The magnitude of discharges is often consistent, forming a ‘’cloud” or ‘’turtle-back” pattern on the PRPD plot. The symmetry arises because the void is bounded by dielectric on both sides, behaving similarly during both positive and negative voltage swings.

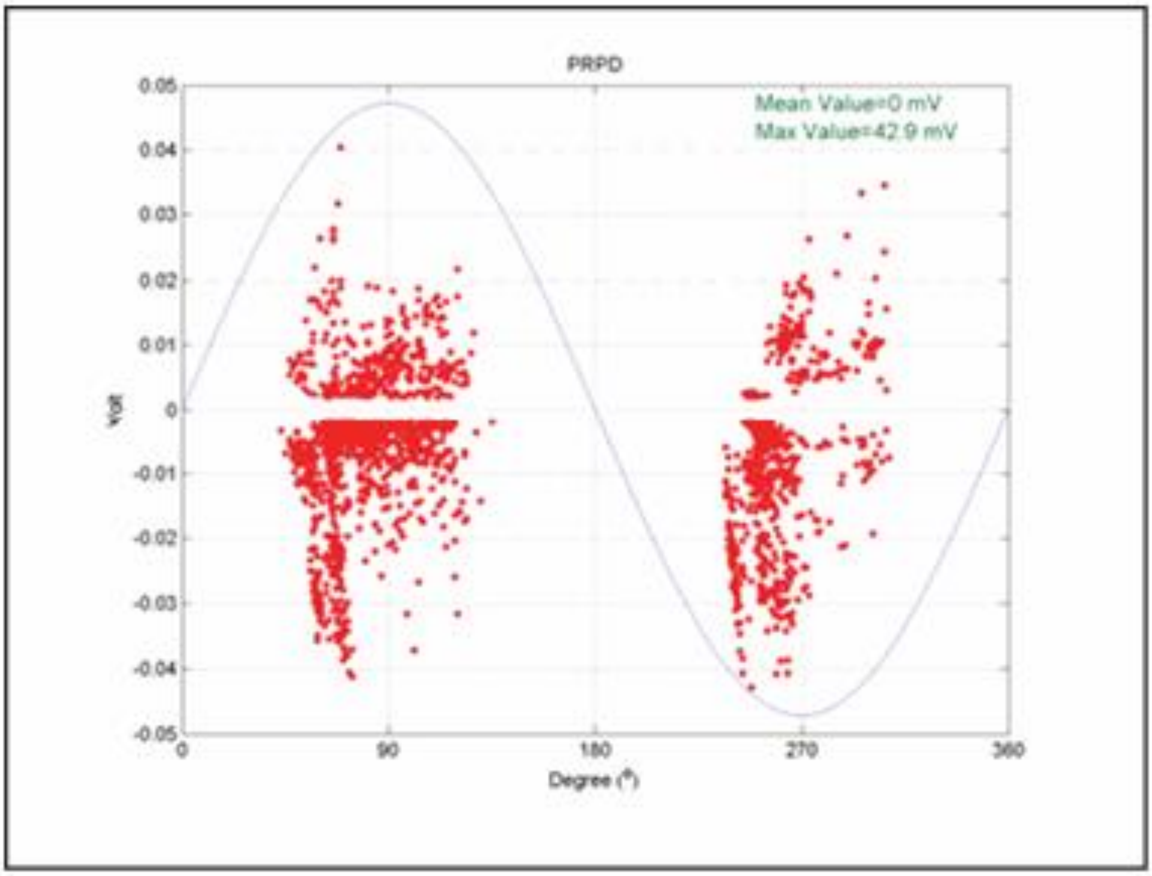

Figure 7 Typical PRPD pattern characteristic of internal void discharge [7].

5.2 Tip (Corona) Discharge

Tip discharge, or corona, occurs at sharp electrode points, protrusions, or floating metal particles where the electric field is highly concentrated. In contrast to internal voids, tip discharge is highly asymmetrical. As shown in Figure 8 (derived from Chou and Chen's experiments), corona of ten manifests as a stable cluster of pulses at the peak of the negative half-cycle (270°). This occurs because when the sharp point is at negative potential (acting as acathode), it readily emits electrons into the surrounding gas/oil via field emission, sustaining a consistent discharge. Positive corona (at 90°) is often more intermittent or requires higher voltage to initiate. This asymmetry is a key diagnostic marker

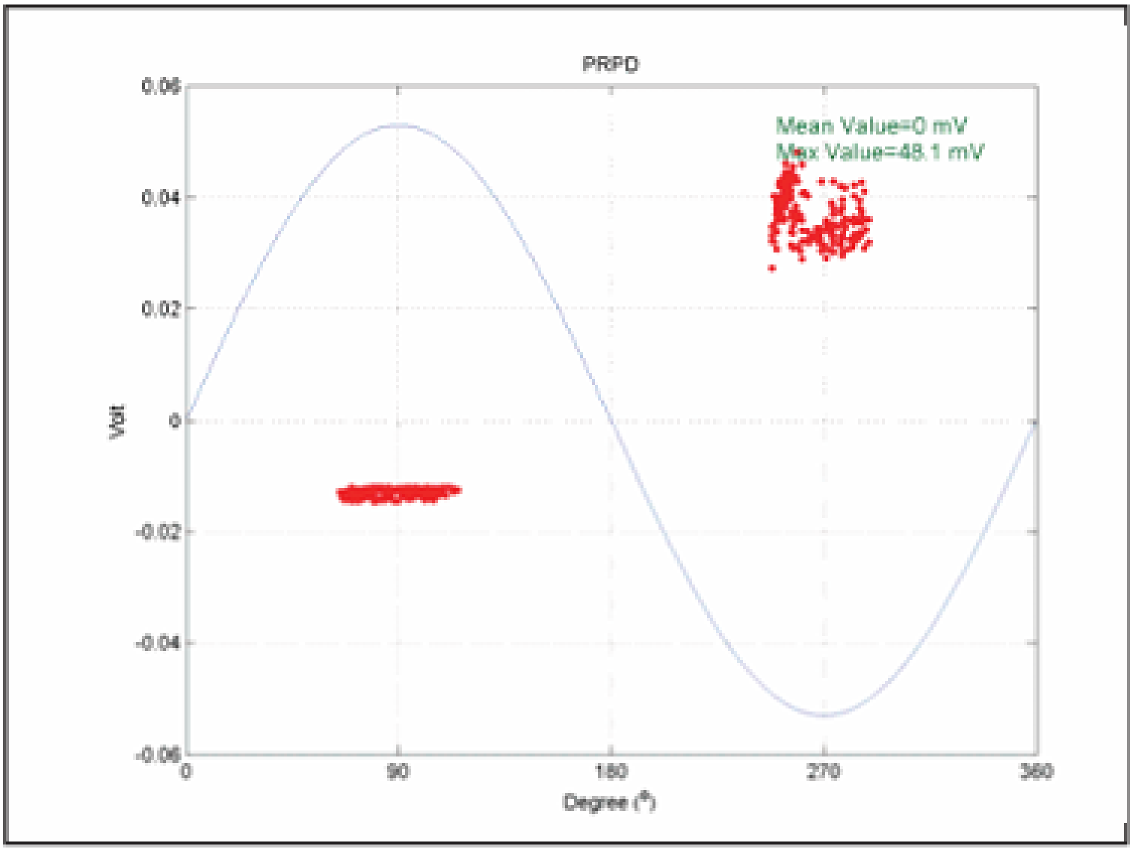

Figure 8 PRPD pattern of tip (corona) discharge [7].

6 Case Studies and Practical Applications

The application of capacitive couplers is governed by international standards to ensure safety and reliability.

6.1 Case Study 1: Rotating Machines Monitoring

IEEE Std1434-2014 [1] provides extensive guidelines for on-line monitoring of motors and generators. In these applications, capacitive couplers (typically 80 pF epoxy-mica capacitors) are permanently installed on the phase terminals or within the iso-phase bus. The standard highlights the importance of differentiating between **Slot Discharge** and **Internal Delamination.** Slot discharge is caused by loose coils vibrating in the stator slot, eroding the semi-conductive coating. This creates a spark gap between the coil surface and the core. The PRPD pattern for slot discharge is often asymmetric, with the positive polarity discharges being larger than the negative ones (depending on the sensor orientation). Conversely, internal de lamination within the ground wall insulation tends to produce symmetric patterns like void discharge.

6.2 Case Study 2: Low-Frequency Testing of Capacitor Banks

For apparatus with very high capacitance, such as large capacitor banks or long cables, energizing the system at 50 Hz requires a power supply with massive kVA rating. Feng et al. [5] explored the use of **Low Frequency (5 Hz)** testing. Their research confirmed that PD inception voltages and pulse repetition rates at 5 Hz are statistically equivalent to those at 50 Hz. This allows for the use of portable, lower-power test sets while still utilizing capacitive couplers for detection.

6.3 Case Study 3: Cable Joint Monitoring

Zhong and Xu [3] demonstrated the efficacy of capacitive couplers for monitoring cable joints, which are the most frequent points of failure in cable systems. They utilized a specialized coupler design where a metallic foil electrode is wrapped around the semi-conducting screen of the cable near the joint. Their experiments on a 46-meter XLPE cable showed that while the high-frequency components of the pulse attenuate rapidly (limiting the ‘’reach’’ of the sensor), placing couplers at joints allows for sensitive local detection. They found that a sensitivity of better than **1pC** could be achieved for defects located within the joint, provided the coupler was installed within a few meters of the defect. This confirms that distributed capacitive sensing is a viable strategy for long transmission lines.

7 Laboratory Experiment: Practical Partial Discharge Measurement with Coupling Capacitor

7.1 Introduction

Following our theoretical modules on high-voltage insulation diagnostics, we participated in a practical laboratory session designed to demonstrate the real-world application of Partial Discharge (PD) measurement. The primary objective of this session was to observe the behaviour of a standard IEC 60270 test circuit, specifically focusing on the role of the coupling capacitor in signal extraction. The professor began the session by explaining that while theoretical models (such as the ‘’abc’’ model) are essential for understanding the physics of discharge, practical measurement requires a sophisticated approach to signal conditioning, impedance matching, and noise suppression. We used a state-of-the-art digital discharge detector to visualize these phenomena.

7.2 Experimental Apparatus and Layout

The experiment was conducted in the high-voltage hall using a setup composed of modular high-voltage components. As we observed in the laboratory layout (Figure 9), the system was dominated by the high-voltage AC test transformer (Hipotronics / Haefely series), which provided the necessary electrical stress. The critical component for our learning objectives was the **Coupling Capacitor (*C_b*)**. In the physical setup, this was a large, distinct unit placed in parallel with the test object.

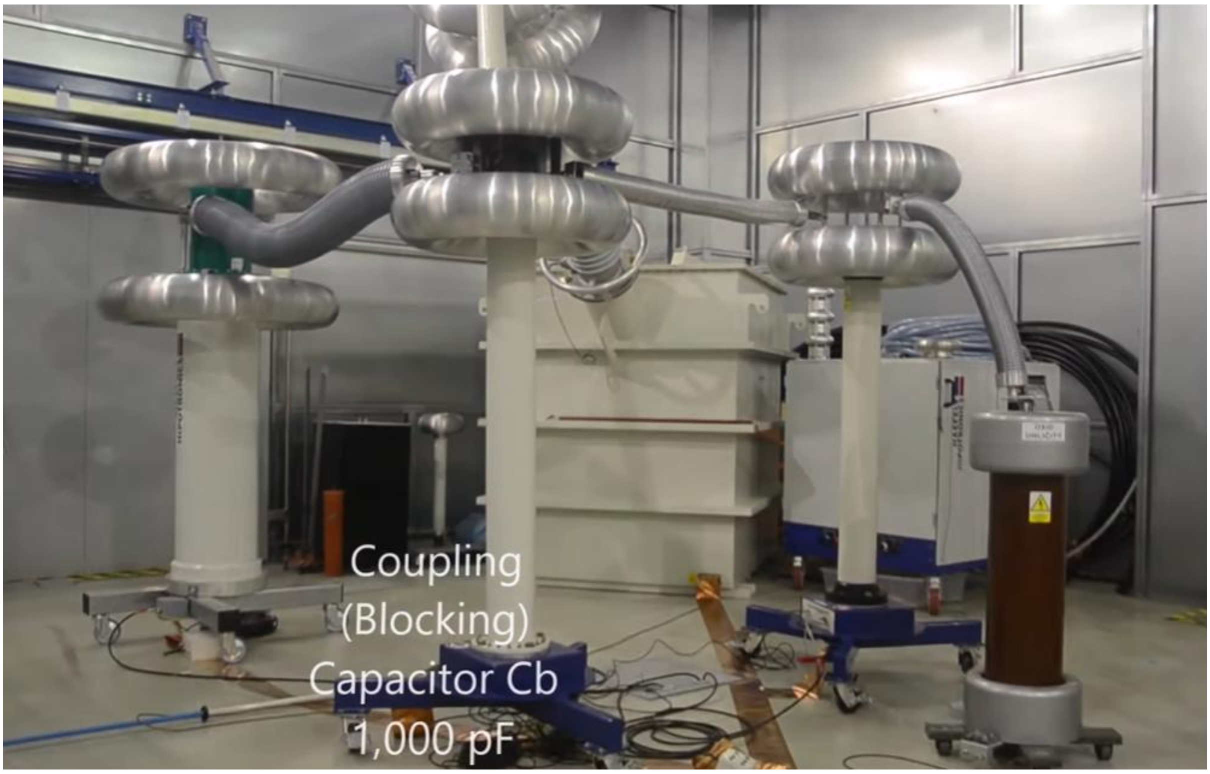

Figure 9 Physical configuration of the high-voltage laboratory test setup.

The apparatus included:

- **High Voltage Source**: A gas-insulated test transformer capable of generating clean AC voltage.
- **Coupling Capacitor (C_b)**: A nominal 1,000 pF capacitor. The professor explained that this component acts as ahigh-pass filter. To the 50Hz power frequency, it presents a very high impedance, effectively blocking the high voltage from reaching the measuring instruments. However, to the nanosecond-scale pulses characteristic of PD (frequencies > 100 kHz), it presents a low impedance path, allowing the signal to be diverted to the detector.
- **Test Object (C_x)**: A standard apacitive load (CIC300) simulating a high-voltage bushing or insulator.

7.3 Circuit Topology and Impedance Matching

To deepen our understanding of the signal path, the professor projected the equivalent circuit diagram of the test setup (Figure 10). We took detailed notes on the specific component values, as these govern the sensitivity and bandwidth of the measurement.

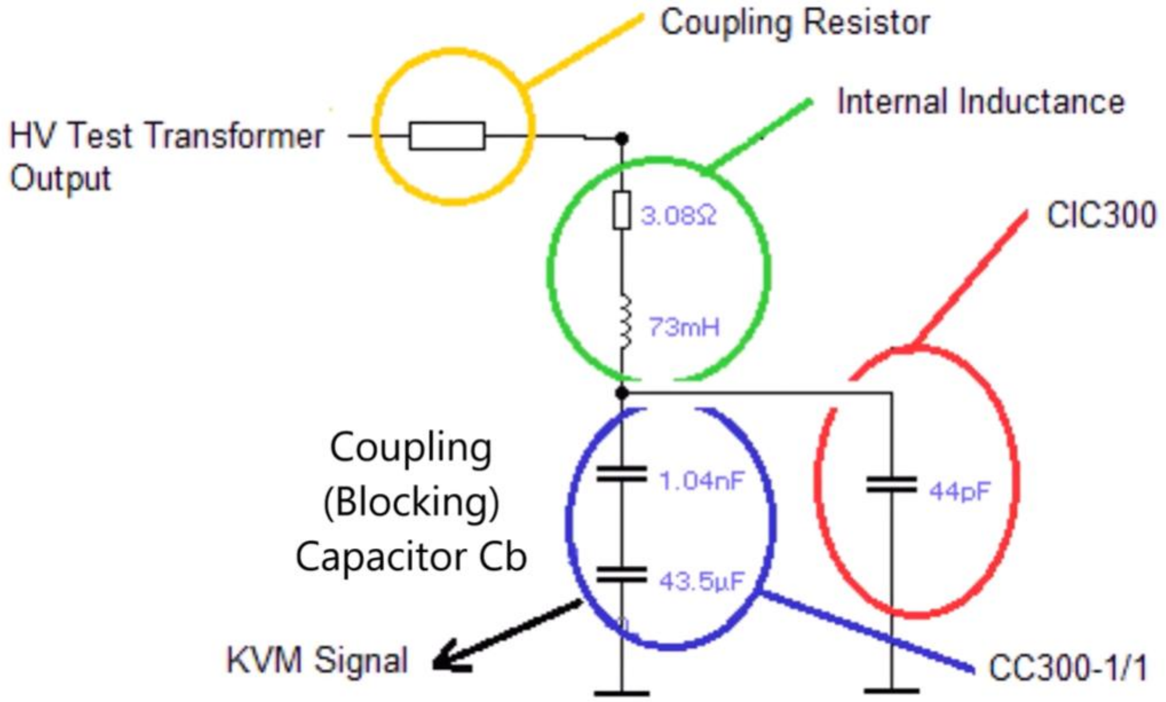

Figure 10 Equivalent circuit diagram of the laboratory setup.

7.4 Theoretical Basis: The Capacitive Model

Before energizing the system, we reviewed the theoretical ''Capacitive Model" that explains how a physical defect inside an insulator manifest as an electrical signal. The professor used a diagram (Figure 11) to illustrate a void trapped within a solid dielectric.

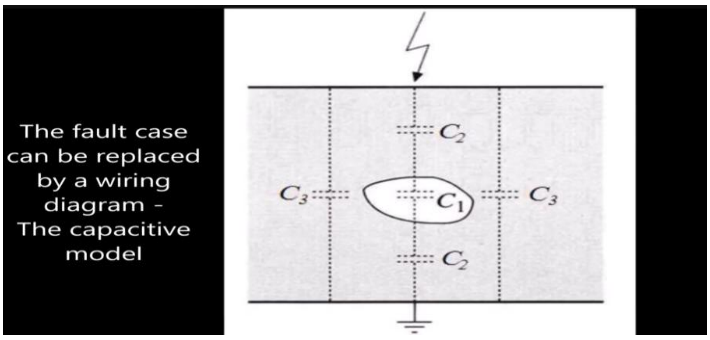

Figure 11 The capacitive model representing an insulation defect.

7.5 Measurement Instrumentation

The data acquisition was performed using a Robinson Instruments DDX-7000 Digital Discharge Detector (Figure 12). This unit represents a significant advancement over older analog oscilloscopes, providing real-time processing, automated calibration, and phase - resolved analysis (PRPD).

Figure 12 The DDX-7000 Digital PD Detector used for signal acquisition

7.6 Calibration and Environmental Noise Assessment

A valid PD measurement cannot be performed without calibration. The measured signal is in volts (at the quadrupole), but the required output is charge. The relationship between the two depends entirely on the circuit capacitances (C_x and C_b). Theprofessorperformedthecalibrationbyinjectingaknownchargepulse(qcal) across the test object. Following calibration, we assessed the electromagnetic environment of the laboratory. High-sensitivity measurements are often plagued by background interference from lighting, radio broadcasts, or power electronics. We viewed the ‘’Amplifier” settings window (Figure 13).

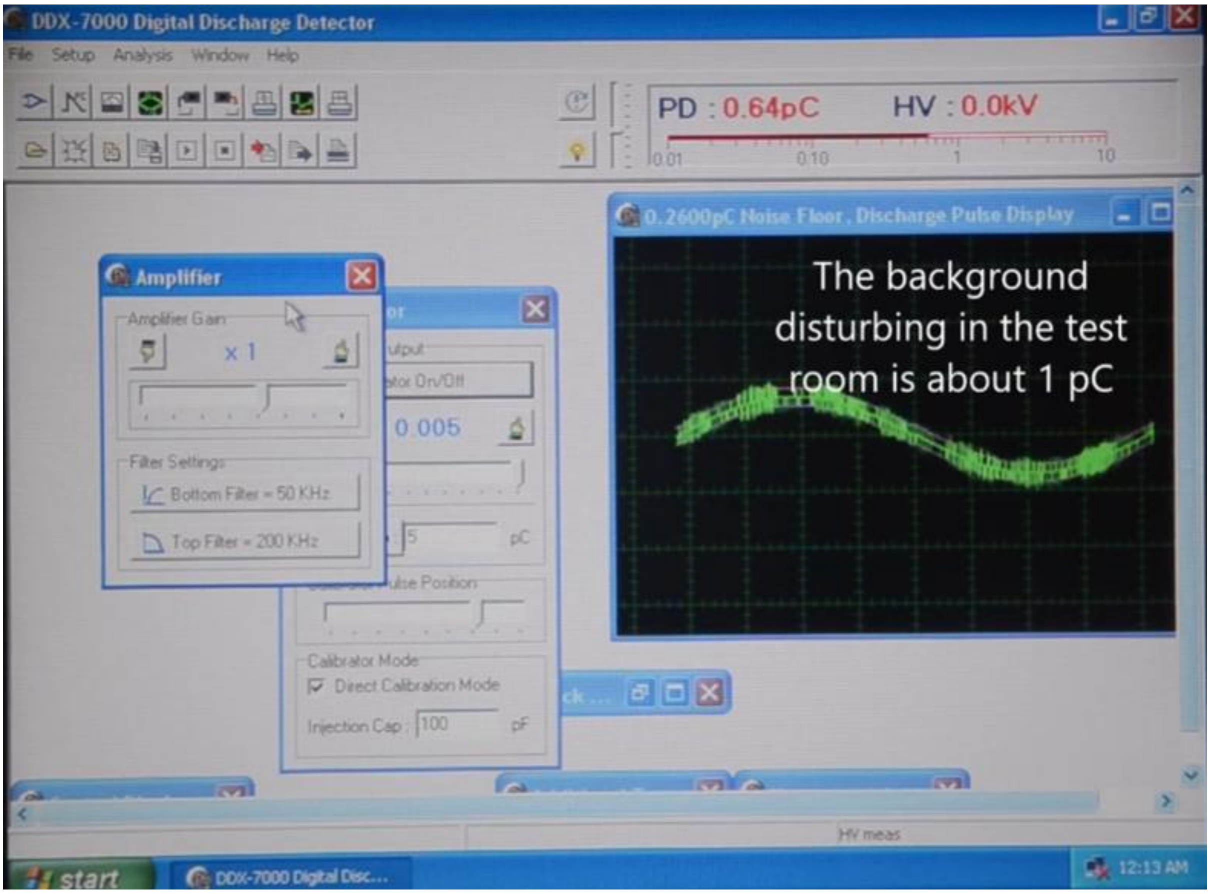

Figure 13 DDX-7000 software interface showing amplifier gain settings.

7.7 Advanced Noise Suppression: Horizontal Gating

To demonstrate how to handle noisier environments, the professor introduced the concept of **Horizontal Discrimination (Gating)**. In many industrial contexts, electrical noise is phase-locked to the power frequency. For example, thyristor switching in exciters often creates noise spikes at specific phase angles. The DDX-7000 allows users to define ‘’gates”—windows in the phase cycle where the detector input is essentially muted.

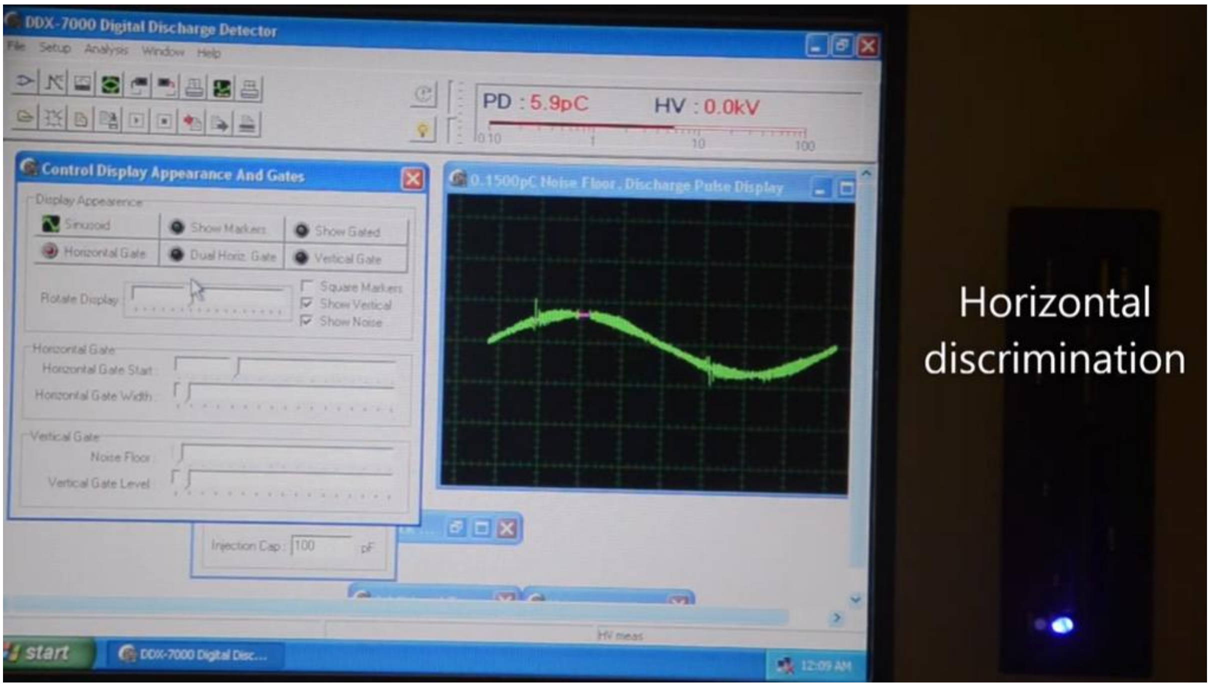

Figure 14 Demonstration of Horizontal Gating.

7.8 Conclusion

This laboratory session successfully demonstrated the practical implementation of the coupling capacitor method. By configuring the test circuit with a 1,000 pF capacitor and using the DDX-7000 for analysis, we were able to detect and quantify low-level discharges with high clarity. The experiment rein forced the importance of the ratio between the coupling capacitor and test object capacitance and provided valuable experience in interpreting phase-resolved PD patterns.

8 Conclusion

The capacitive coupler remains a cornerstone of partial discharge measurement technology. While the fundamental concept—a high-pass filter—is simple, the engineering required to implement it effectively is complex.

1. **Theoretical Accuracy**: A deep understanding of the Dipole Model is necessary to interpret induced charge correctly, moving beyond the limitations of the simple ‘’abc” capacitive model.

2. **Sensor Optimization**: The evolution from simple RC circuits to broad band double-T filters has enabled high-fidelity pulse capture, allowing for integration-based charge measurement with uncertainties as low as 1%.

3. **Noise Rejection**: In field applications, the configuration of the sensors (e.g., balanced bridges) is just as important as the sensors themselves for achieving high sensitivity.

As the power grid ages and expands, the role of capacitive couplers in preventing catastrophic failure through early detection will only grow in importance.

References

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